

Exotic Triangulations and the Octahedral Axiom

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(Co)chain complexes and (co)homology

Mathematicians are often interested in studying the cohomology of some cochain complexes over some abelian category $\mathcal{A}$ (e.g. of modules over some ring).

The category $\mathbf{C}(\mathcal{A})$ is also abelian, which is nice. In particular, we have the notion of short exact sequence, and a short exact sequence of complexes induces a long exact sequence in cohomology

Homotopy category and derived category

Many constructions over morphisms of cochain complexes only depend on the homotopy class, so it makes sense to work in the (smaller) homotopy category

$$\mathbf{K}(\mathcal{A}) := \mathbf{C}(\mathcal{A}) / \simeq .$$

Another standard construction is the derived category $\mathbf{D}(\mathcal{A})$, which inverts the homology isomorphisms.

However, neither of those categories is abelian. But they do have a triangulated structure!

Triangulated categories are a (relatively successful) attempt to axiomatize the properties of $\mathbf{K}(\mathcal{A})$ and $\mathbf{D}(\mathcal{A})$ that are relevant for homological algebra.

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What is a triangulated category? I

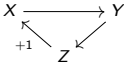
Definition

A *triangulated category* is an additive category $\mathcal{T}$ together with an autoequivalence $\Sigma: \mathcal{T} \rightarrow \mathcal{T}$ and a class of distinguished triangles

$$X \rightarrow Y \rightarrow Z \rightarrow \Sigma X$$

satisfying some axioms (TR1)-(TR4).

Abelian vs triangulated categories

Abelian category	Triangulated category
Is a property of the category	Is some extra structure
Short exact sequences $X \hookrightarrow Y \twoheadrightarrow Z$	Distinguished triangles 
Short exact sequences induce long exact sequences in 'derived functors'	Distinguished triangles induce long exact sequences after applying representable or 'cohomological' functors
First isomorphism theorem: $Z \cong Y/X$	Z only depends on $X \rightarrow Y$ up to (non-canonical) isomorphism

Derived categories are not the only widely used examples of triangulated categories. We could make the following division:

- **Algebraic:** those that are equivalent to the stable category of a Frobenius category $\underline{\mathcal{F}}$ (e.g. $\mathbf{K}(\mathcal{A})$ and $\mathbf{D}(\mathcal{A})$).
- **Topological:** those that are equivalent to the homotopy category of a stable model category $ho\mathcal{M}$.
- **Exotic:** those that are neither algebraic nor topological, i.e. those that admit no enhancement.

Non-exotic = enhanceable

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There are only a few known examples of exotic triangulated categories:

- Muro, Schwede and Strickland [MSS07] gave an exotic structure to the category of finitely generated projective modules over $\mathbb{Z}/4\mathbb{Z}$.
- Rizzardo and Van den Bergh [RV20] gave a class of exotic examples which are linear over fields of characteristic zero.

We have found a third class of examples which are linear over algebraically closed fields of any characteristic.

Our example: Underlying category

We consider the preprojective algebra of type L_2 over an algebraically closed field k . This is the path algebra

$$P(L_2) = k \left(b \curvearrowright 1 \begin{array}{c} \xrightarrow{a_1} \\ \xleftarrow{a_2} \end{array} 2 \right) / (b^2 + a_2 a_1, a_1 a_2).$$

If you are not familiar with path algebras, just think of it as the free algebra generated by elements

$$\{e_1, e_2, a_1, a_2, b\}$$

with the relations

$$\begin{aligned} e_1^2 &= e_1, & e_2^2 &= e_2, & e_1 e_2 &= 0 = e_2 e_1, \\ e_2 a_1 &= a_1 = a_1 e_1, & e_1 a_2 &= a_2 = a_2 e_2, & e_1 b &= b = b e_1, \\ b^2 + a_2 a_1 &= 0, & a_1 a_2 &= 0. \end{aligned}$$

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Our example: Underlying category

We consider the category of finitely generated modules over $P(L_2)$. This category is equivalent to the category of representations of $P(L_2)$, which has:

- Objects: Diagrams of f.d. vector spaces of the form

$$B \hookrightarrow V_1 \begin{matrix} \xrightarrow{A_1} \\ \xleftarrow{A_2} \end{matrix} V_2$$

which satisfy the relations $B^2 + A_2A_1 = 0$ and $A_1A_2 = 0$.

- Morphisms: Commutative diagrams of the form

$$\begin{array}{ccccc} B \hookrightarrow V_1 & \begin{matrix} \xrightarrow{A_1} \\ \xleftarrow{A_2} \end{matrix} & V_2 \\ \downarrow f_1 & & \downarrow f_2 \\ C \hookrightarrow W_1 & \begin{matrix} \xrightarrow{D_1} \\ \xleftarrow{D_2} \end{matrix} & W_2 \end{array}$$

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Our example: Underlying category

The underlying category of our example is the category of finitely generated projective modules over $P(L_2)$, whose objects are direct sums of the following two indecomposable representations:

$$P^1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \hookrightarrow k^4 \begin{array}{c} \xrightarrow{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}} \\ \xleftarrow{\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}} \end{array} k^2$$

$$P^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \hookrightarrow k^2 \begin{array}{c} \xrightarrow{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}} \\ \xleftarrow{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}} \end{array} k^2$$

Our example: Shift functor

Our shift functor is given by twisting by the automorphism

$$\sigma: P(L_2) \rightarrow P(L_2), \quad a_1 \mapsto -a_1, \quad a_2 \mapsto -a_2, \quad b \mapsto -b - b^3.$$

Explicitly, it is the restriction of the functor

$$(-)_\sigma: \text{mod}(P(L_2)) \rightarrow \text{mod}(P(L_2))$$

defined by:

$$B \curvearrowright V_1 \begin{matrix} \xrightarrow{A_1} \\ \xleftarrow{A_2} \end{matrix} V_2 \mapsto -B-B^3 \curvearrowright V_1 \begin{matrix} \xrightarrow{-A_1} \\ \xleftarrow{-A_2} \end{matrix} V_2$$

$$(f_1, f_2) \mapsto (f_1, f_2).$$

Our example: Triangles

The distinguished triangles are sums of shifts of the following ones:

$$0 \rightarrow P^1 \xrightarrow{id} P^1 \rightarrow 0,$$

$$0 \rightarrow P^2 \xrightarrow{id} P^2 \rightarrow 0,$$

$$P_\sigma^1 \xrightarrow{b^3} P^1 \xrightarrow{\begin{pmatrix} b \\ -a_2 \end{pmatrix}} P^1 \oplus P^2 \xrightarrow{\begin{pmatrix} b & a_1 \end{pmatrix}} P^1,$$

$$P_\sigma^1 \xrightarrow{a_2 b} P^2 \xrightarrow{b a_1} P^1 \xrightarrow{b} P^1,$$

$$P_\sigma^1 \oplus P_\sigma^2 \xrightarrow{\begin{pmatrix} -b^2 & -b a_1 \\ a_2 b & 0 \end{pmatrix}} P^1 \oplus P^2 \xrightarrow{\begin{pmatrix} -b^2 & 0 \\ b & a_1 \end{pmatrix}} P^1 \oplus P^1 \xrightarrow{\begin{pmatrix} b & b^2 \\ -a_2 & 0 \end{pmatrix}} P^1 \oplus P^2,$$

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$$P_\sigma^2 \xrightarrow{-a_2 b a_1} P^2 \xrightarrow{a_1} P^1 \xrightarrow{a_2} P^2.$$

Idea of the construction

The idea comes from the following result:

Theorem (F. Muro, 2022, [Mur22])

Let Λ be a basic connected non-separable twisted-3-periodic f.d. algebra over a perfect field k . Then, there is an automorphism $\tau \in \text{Aut}(\Lambda)$ such that twisting by τ is essentially the unique shift functor $\Sigma = (-)_{\tau}: \text{proj}(\Lambda) \rightarrow \text{proj}(\Lambda)$ that makes $(\text{proj}(\Lambda), \Sigma)$ admit an algebraic triangulated structure.

So, the idea is to modify the automorphism τ into another automorphism σ such that $(-)_{\tau}$ is not isomorphic to $(-)_{\sigma}$ but in a way that $\text{proj}(\Lambda)$ still admits a pretriangulated structure with shift $(-)_{\sigma}$.

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In particular, the modification comes from composing with another automorphism κ which is a representative for a nontrivial element of the kernel of

$$Out(\Lambda) \rightarrow \underline{Pic}(\Lambda),$$

where

$$Out(\Lambda) = Aut(\Lambda)/Inn(\Lambda),$$

$$\underline{Pic}(\Lambda) := \{ \text{Triangulated self-equivalences of } \underline{mod}(\Lambda) \} / \cong_{tr},$$

and the map is given by $[\varphi] \mapsto [(-)_{\varphi}]$.

Idea of the construction

Theorem

Let Λ and τ be as in the previous theorem. Then, any nontrivial element κ in the kernel of $\text{Out}(\Lambda) \rightarrow \underline{\text{Pic}}(\Lambda)$ gives rise to a non-algebraic pretriangulated structure on $(\text{proj}(\Lambda), (-)_{\kappa \circ \tau})$.

Furthermore, our particular example can also be shown to be non-topological.

Bonus: The octahedral axiom

The octahedral axiom (TR4) is the axiom of triangulated categories which is an analogue of the second isomorphism theorem in abelian categories:

$$A \subseteq B \subseteq C \quad \rightsquigarrow \quad \frac{C/A}{B/A} \cong C/B.$$

Open problem: There is no known example of a pretriangulated category that does not satisfy the octahedral axiom, nor has it been proven to be a consequence of the other axioms (TR1)-(TR3).

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What we have tried

- Recover it from the octahedral axiom on the algebraic structure.
- Adapt the proof of the algebraic case.
- Try to disprove it by finding a counterexample by computer. We have not been able to find any, but the search is slow.
- Use a (computable) sufficient condition given in [Mur20] in terms of the differential of a spectral sequence. It turns out that it is not satisfied. It is also the first example where this condition has been checked with a negative result.

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- Give an equivalent condition for the octahedral axiom in terms of a secondary cohomology operation. We have been able to get this, but it is not clear whether or not it helps.
- **Classifying initial data:** We can try to classify all the possible initial data and check the axiom for each possible case. This is equivalent to classifying all morphisms of $\underline{mod}(\Lambda)$. We don't know how to do this, but it may be a path to follow.

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- [MSS07] Fernando Muro, Stefan Schwede, and Neil Strickland. “Triangulated Categories without Models”. In: *Inventiones Mathematicae* 170.2 (2007), pp. 231–241. ISSN: 0020-9910. DOI: 10.1007/s00222-007-0061-2.
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Thank you for your attention!